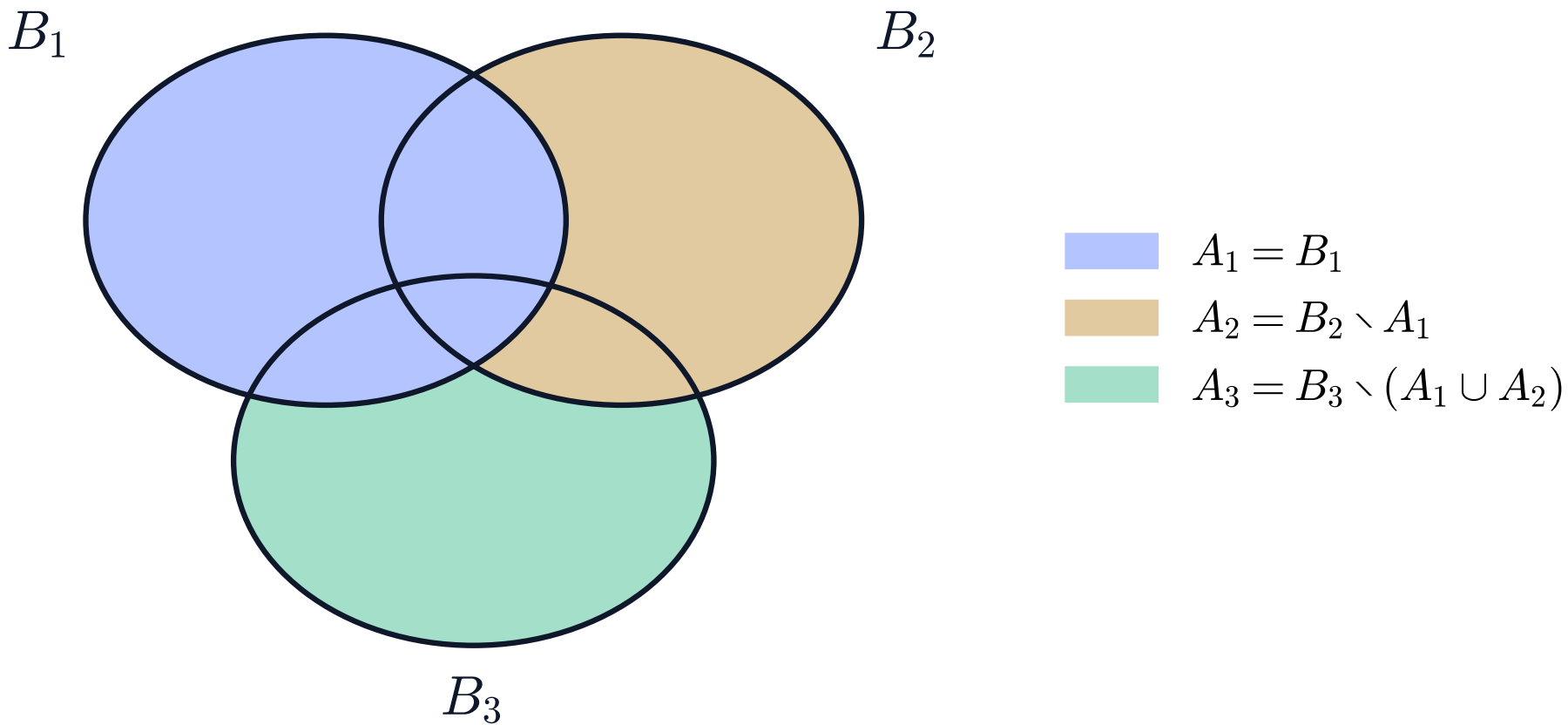
B_1
B_2
B_3
$A_1 = B_1$
$A_2 = B_2 \setminus A_1$
$A_3 = B_3 \setminus (A_1 \cup A_2)$